DDPM Exam 2 — Comprehensive Cheat Sheet (Modules 4 & 5)

DDPM Exam 2 Cheat Sheet (Modules 4, 5 & 6)

This document provides a comprehensive, source-faithful module-by-module review across all 15 syllabus topics for Exam 2 of the Data-Driven Product Management programme.

> [!IMPORTANT]
>
> **Operational Parameters & Scope Status**
>
> - **Confirmed Operational Parameters**: 45 questions | 60-minute duration (3:30 PM–4:30 PM IST, Sun 16 Aug) | No negative marking (answer 100% of questions).
>
> - **Source-Missing Material**: Topics with missing local class assets are marked as **externally supplemented; instructor-specific coverage unresolved**.

Module 4: Building Product Market Strategy

Topic M04-T01: Marketing Strategy and Marketing Plan (`[class-confirmed]`)

1. Strategic Context & Expansion Options (Natureview Farm Case)

Natureview Farm, an organic yogurt manufacturer with $13.0M in 1999 revenues (`DDPM_Organized/03_Sessions/S16_2026-05-31_Marketing_strategy_Prof_Anita_Goyal/Slides/Natureview EPDDPM-1.pdf`, Exhibit 1), evaluated three growth options to achieve $20.0M in revenue by end of 2001:

- **Option 1: 8 oz SKU Expansion into Supermarkets (East & Central)**
- Target: Enter conventional supermarket channel with 6 SKUs of 8 oz yogurt.
- Financial Terms: One-time slotting fee of $10,000 per SKU per store chain (`DDPM_Organized/03_Sessions/S16_2026-05-31_Marketing_strategy_Prof_Anita_Goyal/Slides/Natureview EPDDPM-1.pdf`, Slide 5).
- Trade Promotions: 15% off-invoice promotion + co-op advertising allowances.
- Broker Commission: 5% of manufacturer sales.
- **Option 2: 32 oz SKU National Expansion into Supermarkets**
- Target: Enter conventional supermarket channel with 4 SKUs of 32 oz yogurt.
- Economics: Lower slotting expenses relative to Option 1 due to fewer SKUs, lower promotional intensity (marketing expenses ~10% of 8-oz expansion) (`DDPM_Organized/03_Sessions/S16_2026-05-31_Marketing_strategy_Prof_Anita_Goyal/Slides/Natureview EPDDPM-1.pdf`, Slide 8).
- **Option 3: 4 oz Multipack in Natural Foods Channel**
- Target: Expand within existing natural foods channel where Natureview held a 24% market share (`DDPM_Organized/03_Sessions/S16_2026-05-31_Marketing_strategy_Prof_Anita_Goyal/Slides/Natureview EPDDPM-1.pdf`, Exhibit 5).
- Strategic Fit: Capitalized on 15% 5-year CAGR in natural foods channel (`DDPM_Organized/03_Sessions/S16_2026-05-31_Marketing_strategy_Prof_Anita_Goyal/Slides/Natureview EPDDPM-1.pdf`, Slide 9).

2. Baseline Financials (1999 Income Statement, Exhibit 1)

- Revenues: $13,000,000 (100%)
- Cost of Goods Sold (COGS): $8,190,000 (63%)

- Gross Profit: $4,810,000 (37%)
- Admin & Freight: $2,210,000 (17%)
- Sales Expenses: $1,560,000 (12%)
- Marketing Expenses: $390,000 (3%)

3. Key Formulas

$$\text{Manufacturer Net Price} = \text{Retail Price} \times (1 - \text{Retail Margin \%}) \times (1 - \text{Wholesaler Margin \%})$$

$$\text{Break-Even Volume (Units)} = \frac{\text{Fixed Marketing Expenses (Slotting Fees + Ad Spend)}}{\text{Manufacturer Contribution Margin per Unit}}$$

Topic M04-T02: Lead Generation and Decoding Customer Journey (`[class-confirmed]`)

1. SERVQUAL Model (5 Service Quality Dimensions - RATER)

Parasuraman, Zeithaml & Berry (1988) evaluate service quality gap ($\text{Perception} - \text{Expectation}$) across 5 dimensions (`DDPM_Organized/03_Sessions/S17_2026-06-07_Lead_Generation_and_Customer_Journey_Prof_Prem_Dewani/Slides/Servequal-1.pdf`, Page 4):

1. **Reliability**: Ability to perform promised service dependably and accurately.
2. **Assurance**: Employee knowledge, courtesy, and ability to inspire trust and confidence.
3. **Tangibles**: Physical facilities, equipment, staff appearance, and digital UI touchpoints.
4. **Empathy**: Caring, individualized attention provided to customers.
5. **Responsiveness**: Willingness to help customers and provide prompt service.

2. Commitment-Trust Theory of Relationship Marketing

Morgan & Hunt (1994) identify **Relationship Commitment** and **Trust** as Key Mediating Variables (KMVs) (`DDPM_Organized/03_Sessions/S17_2026-06-07_Lead_Generation_and_Customer_Journey_Prof_Prem_Dewani/Readings/Commitment and Trust theory of Relationship Marketing.pdf`, Page 22):

- **Trust**: Confidence in an exchange partner's reliability and integrity. Driven by shared values, communication, and non-opportunistic behavior.
- **Commitment**: Enduring desire to maintain a valued relationship. Reduces termination costs and increases customer retention.

3. Mental Accounting & Consumer Choice (Thaler)

- **Acquisition Utility**: $U_{acq} = v(p) - p$ (Value of goods received vs outlay).
- **Transaction Utility**: $U_{trans} = v(p^*) - p$ (Perceived deal quality relative to reference price p^*).
- **Four Principles of Mental Accounting**:

1. Segregate gains (multiple small pleasant surprises).
2. Integrate losses (single lump sum bill).
3. Integrate small losses with larger gains.
4. Segregate small gains from large losses (silver lining effect).

Topic M04-T03: Concept Testing and B2C Product Development (`[class-confirmed]` / `[credible-secondary]`)

1. Concept Testing Metrics & 5-Point Purchase Intent Scale

Concept testing presents a structured concept statement to target consumers (`DDPM_Organized/03_Sessions/S18_2026-06-14_Concept_Testing_Prof_Anita_Goyal/Recordings/1st half.m4a`):

- Scale: 5 = Definitely Would Buy, 4 = Probably Would Buy, 3 = Might/Might Not Buy, 2 = Probably Would Not Buy, 1 = Definitely Would Not Buy.
- **Top 2 Box (T2B)** = % Definitely Buy + % Probably Buy.

2. Purchase Intent Discounting (Nielsen BASES Model)

Survey purchase intent overstates actual market behavior. Standard calibration discounting rules apply:

$$\text{Calibrated Trial \%} = (0.80 \times \text{Top Box \%}) + (0.30 \times \text{Second Box \%})$$

3. ATAR Adoption Model (Urban & Hauser, 1993)

Forecasts market penetration and sales volume for new B2C products:

- **Awareness (A)**: % of target market aware of the product.
- **Trial (T)**: % of aware consumers making an initial trial purchase.
- **Availability (A)**: % distribution coverage (ACV - All Commodity Volume).
- **Repeat (R)**: % of trial buyers making repeat purchases.

$$\text{Market Share \%} = A \times T \times A \times R$$

Topic M04-T04: Media Mix, Messaging, Packaging, and Bundling (`[class-confirmed]` / `[credible-secondary]`)

1. Integrated Marketing Communications (IMC) & Media Classification

- **Above-The-Line (ATL)**: Mass media (TV, Radio, Print) to build mass awareness and brand equity.
- **Below-The-Line (BTL)**: Direct promotions, point-of-sale displays, trade allowances driving immediate conversion.
- **Through-The-Line (TTL)**: Integrated digital marketing, social media, search, and influencer campaigns.

2. Strategic Functions of Packaging (Kotler & Keller, 2015; Robertson, 2013)

1. **Functional & Protective**: Preserves product integrity, safety, and shelf-life during logistics.

2. **Informational & Regulatory**: Displays ingredient statements, nutritional facts, usage instructions, compliance labels, and barcodes.

3. **Perceptual & Branding ("Silent Salesman")**: Communicates brand identity, positioning, and visual differentiation at point of sale (`DDPM_Organized/03_Sessions/S19_2026-06-21_Media_Mix_and_Packaging_Prof_Anita_Goyal/Screenshots/Screenshot 2026-06-21 at 16.09.12.png`).

3. Product Bundling Strategies (Stremersch & Tellis, 2002)

- **Pure Bundling**: Products sold exclusively as a bundle; components cannot be purchased individually.
- **Mixed Bundling**: Products offered both individually and as a discounted bundle.
- **Economic Rationale**: Captures consumer surplus across heterogeneous valuation segments.

Topic M04-T05: Managing Channel Partners and Affiliates (`[source-missing] — externally supplemented; instructor-specific coverage unresolved`)

1. Marketing Channel Architecture & Power (Coughlan et al., 2006)

- **Channel Conflict**:
- *Vertical Conflict*: Conflict between different channel levels (e.g., manufacturer selling D2C cheaper than retail partner MSRP).
- *Horizontal Conflict*: Conflict between channel members at the same level (e.g., price war between two regional distributors).
- **French & Raven's 5 Bases of Channel Power**: Coercive, Reward, Legitimate, Expert, Referent power.

2. Affiliate Marketing Payment Models

- **Pay-Per-Click (PPC)**: Affiliate compensated per traffic click.
- **Pay-Per-Lead (PPL)**: Affiliate compensated per qualified user lead/registration.
- **Pay-Per-Sale (PPS)**: Affiliate compensated per completed transaction.

3. Marketing Attribution Platform Specifications

- **Google Analytics 4 (GA4)** (`[official-external: GA4 Specification]`):
- Retired legacy rules-based models (First Touch, Linear, Time-Decay, rules-based Position-Based) in mid-2023.
- Supports 3 models: **Data-Driven Attribution (DDA)** (machine learning sequence analysis), **Paid and Organic Last Click**, and **Google Paid Channels Last Click**.
- **Adobe Marketo Measure (Enterprise B2B Multi-Touch)** (`[official-external: Adobe Specification]`):
- **U-Shaped**: Allocates **50%** credit to First Touch, **50%** to Lead Creation Touch, and **0%** to intermediary touches.
- **W-Shaped**: Allocates **30%** to First Touch, **30%** to Lead Creation Touch, **30%** to Opportunity Creation Touch, and **10%** split equally across middle touches.

Topic M04-T06: Pricing Models (`[class-confirmed]`)

1. Strategic Pricing Paradigms

- **Cost-Plus Pricing**: Fixed markup added to cost; simple but ignores customer WTP.
- **Value-Based Pricing**: Prices set based on customer Willingness-to-Pay (WTP) and Economic Value to Customer (EVC).
- **Dynamic Pricing**: Algorithmic price adjustments based on real-time demand, time, and inventory constraints.

2. Tiers of Price Discrimination

1. **First-Degree (Personalized)**: Charging each customer their exact maximum WTP.
2. **Second-Degree (Versioning / Quantity Discounting)**: Self-selecting volume/package options (e.g., single game tickets vs 5-game vs 20-game vs 38-game full season) (`DDPM_Organized/03_Sessions/S21_2026-07-05_Pricing_models_Prof_Prem_Dewani/Slides/The Springfield.pdf`).
3. **Third-Degree (Segmented)**: Charging different prices to distinct customer segments (e.g., stadium grandstand vs bleacher seating).

3. Springfield Nor'easters Case Analysis

- **Fixed Annual Operating Expenses**: $1,961,379 (`DDPM_Organized/03_Sessions/S21_2026-07-05_Pricing_models_Prof_Prem_Dewani/Slides/The Springfield.pdf`, Exhibit 1).
- **Concession Expectations**: 45% expect to spend $6–$10, 36% expect $11–$15 per person per game (`DDPM_Organized/03_Sessions/S21_2026-07-05_Pricing_models_Prof_Prem_Dewani/Slides/The Springfield.pdf`, Exhibit 5 Q13).
- **Optimal Price Formula**:

$$E_p = \frac{\% \Delta Q}{\% \Delta P}, \quad \text{Optimal Price} = \frac{MC}{1 + (1 / E_p)}$$

Module 5: Defining User Experiences - Tools

Topic M05-T01: AI in Customer Experience and Personalisation (`[source-missing] — externally supplemented; instructor-specific coverage unresolved`)

1. Personalization Architecture

- **Rule-Based vs AI Personalization**: Static demographic rules vs dynamic machine learning engines evaluating behavioral clickstreams and real-time context.
- **Next-Best-Action (NBA) Engine**: Algorithmic decisioning to serve the optimal offer, message, or support prompt in real time.

2. Recommendation Algorithms (Aggarwal, 2016)

- **Collaborative Filtering**:
- *User-Based*: Recommends items preferred by users with similar rating histories.
- *Item-Based*: Recommends items co-purchased or co-viewed with target item (e.g., "Customers who bought this also bought...").
- **Content-Based Filtering**: Recommends items matching features of items previously preferred by user.
- **Hybrid Systems**: Combines collaborative and content-based filtering to overcome cold-start issues.

3. Conversational AI & RAG Support

- NLU intent recognition, automated ticket triage, and Retrieval-Augmented Generation (RAG) over enterprise knowledge bases.

Topic M05-T02: Customer Engagement and Retention Strategies (`[source-missing] — externally supplemented; instructor-specific coverage unresolved`)

1. Core Retention Metrics (Fader & Hardie, 2007)

$$\text{Customer Churn Rate (CR)} = \frac{\text{Customers Lost During Period}}{\text{Customers at Start of Period}}$$

$$\text{Customer Retention Rate (CRR)} = \frac{\text{End Customers} - \text{New Customers}}{\text{Start Customers}} \times 100$$

$$\text{Customer Lifetime Value (CLV)} = \frac{\text{ARPU} \times \text{Gross Margin \%}}{\text{Monthly Churn Rate}}$$

$$\text{Net Promoter Score (NPS)} = \% \text{Promoters (9-10)} - \% \text{Detractors (0-6)}$$

2. Cohort Analysis & Retention Curves

- Tracking user cohorts over time ($T_0, T_1, T_2, \dots$).
- Retention curve flattening horizontally indicates true Product-Market Fit (PMF); downward sloping curve to zero indicates lack of PMF.

3. RFM Customer Segmentation

- **Recency (R)**, **Frequency (F)**, **Monetary (M)** value.
- Clusters: Champions (High R, High F, High M), At-Risk (Low R, High F, High M — primary win-back targets), Hibernating (Low R, Low F, Low M).

Topic M05-T03: Cyber Risk Management and Trust (`[source-missing] — externally supplemented; instructor-specific coverage unresolved`)

1. NIST Cybersecurity Framework (NIST CSF v2.0 Core Functions) (`[official-external: NIST CSF v2.0]`)

NIST CSF v2.0 establishes 6 core functions:

1. **Govern (GV)**: Establishes organizational cybersecurity risk strategy, policies, roles, and supply chain risk oversight (newly added 6th function in CSF v2.0).
2. **Identify (ID)**: Asset management, risk assessment, operational system discovery.
3. **Protect (PR)**: Access control, identity management, staff training, data encryption.
4. **Detect (DE)**: Continuous monitoring, anomaly detection, intrusion detection.
5. **Respond (RS)**: Incident triage, containment, mitigation, forensics.
6. **Recover (RC)**: Restoration of services post-incident and resilience planning.

2. Privacy by Design (PbD) & Zero Trust Architecture (ZTA)

- **Privacy by Design (7 Principles, Cavoukian 2009)**: Proactive not reactive, Privacy as default, Privacy embedded into design, Full functionality, End-to-end security, Visibility/transparency, Respect for user privacy.

- **Zero Trust Architecture (NIST SP 800-207)**: "Never trust, always verify." Guiding principles: explicit verification, least privilege, assume breach posture.

Topic M05-T04: Tableau Session (`[source-missing] — externally supplemented; instructor-specific coverage unresolved`)

1. Tableau Data Architecture & Visual Encodings (`[official-external: Tableau Help]`)

- **Dimensions vs Measures**: Dimensions are categorical/qualitative (slice data); Measures are numerical/quantitative (aggregated via `SUM`, `AVG`).
- **Discrete vs Continuous**: Discrete (Blue pills) creates distinct headers; Continuous (Green pills) creates continuous axes.

2. Level of Detail (LOD) Expressions

1. **FIXED LOD**: `{ FIXED [Region] : SUM([Sales]) }` — Computes aggregate at specified dimension granularity, ignoring view dimensions. Evaluated *before* dimension filters.

2. **INCLUDE LOD**: `{ INCLUDE [Customer ID] : AVG([Sales]) }` — Computes aggregate including specified dimensions in addition to view dimensions. Evaluated *after* dimension filters.

3. **EXCLUDE LOD**: `{ EXCLUDE [Category] : SUM([Sales]) }` — Computes aggregate excluding specified view dimensions. Evaluated *after* dimension filters.

3. Tableau Order of Operations (Filter Pipeline Execution)

1. Extract Filters $\rightarrow$ 2. Data Source Filters $\rightarrow$ 3. Context Filters $\rightarrow$ **4. FIXED LOD Calculations** $\rightarrow$ 5. Dimension Filters $\rightarrow$ **6. INCLUDE/EXCLUDE LOD Calculations** $\rightarrow$ 7. Measure Filters $\rightarrow$ 8. Table Calculations.
